

Shubha Barman

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SUMMARY

Versatile engineer with skills in software development, machine learning, and data science. Experienced in Python, building scalable applications, and developing AI/ML and Generative AI solutions including LLM-based systems and RAG. Passionate about solving real-world problems using data-driven and intelligent technologies.

EDUCATION

Ramkrishna Mahato Government Engineering College

2020 - 2024

BTech, Computer science and Engineering - 8.19 (CGPA)

Siliguri Baradakanta Vidyapith(H.S)

Madhyamik(10th)

Higher secondary(12th)

PROJECTS

Large Language Model (LLM) Implementation | Tech Stack: Python, Pytorch, Jupyter notebook

- Built a custom Transformer-based LLM using PyTorch for human-like text generation.
- Developed the full training pipeline, including tokenization and optimization strategies.

DeepSeek From Scratch

- Familiar with advanced Transformer components including **Multi-Head Latent Attention (MLA)**, **Mixture of Experts (MoE)**, and **Rotary Positional Embeddings (RoPE)**.

Lovable Clone Assistant | Tech Stack: Python, OpenAI API, LangGraph, Streamlit [↗](#)

- AI-powered coding assistant built with LangGraph
- Transforms natural language requests into complete, working projects using real developer workflows

Image to Story to Audio Generator | Tech Stack: Python, Hugging Face Transformers, Streamlit [↗](#)

- Developed a cutting-edge project leveraging Hugging Face models for seamless image-to-text, text-to-story, and story-to-speech audio transformations.

Meeting-to-CRM Agent on Twenty CRM | Tech Stack: Python, FastAPI, LLMs, CRM Integration, Microsoft Meeting API

- Built an AI automation agent that syncs meeting data with Twenty CRM by generating summaries, extracting tasks, and updating customer records automatically using Python and FastAPI.

Math Agent | Tech Stack: Guardrails, RAG, Vector DB, MCP, FastAPI [↗](#)

- Built an **AI-powered Math Agent** using **Retrieval-Augmented Generation (RAG)** and **Vector Database** to retrieve relevant mathematical knowledge and generate accurate solutions.

Chat-With-Pdf | Tech Stack: Amazon Bedrock, S3, Docker, RAG, FAISS, Langchain, Streamlit [↗](#)

- Chat-With-PDF is a web app that lets users interact with PDFs via natural language.
- It uses Amazon Bedrock, RAG, and LangChain for intelligent PDF interaction.

SKILLS

- **Programming & Frameworks:** Python, SQL, C++/C, PyTorch, TensorFlow, Keras, Scikit-learn, Streamlit, React.js, TypeScript, FastAPI
- **LLM Orchestration Frameworks:** LangChain, LangGraph, Model Context Protocol (MCP), Hugging Face
- **CI/CD & DevOps:** GitHub Actions, Docker, Kubernetes, Git, GitHub, AWS
- **Research Exposure (Transformer Architectures):** Vision Transformer (ViT), DiET Transformer-, BERT (Bidirectional Encoder Representations from Transformers), Swin Transformer (Shifted Window Transformer)
- **Vector Databases:** FAISS, Chroma, Pinecone
- **Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, Neural Networks, CNNs, RNNs, Transformers, Natural Language Processing (NLP), Computer Vision (image classification, object detection)
- **Data Handling & Analytics:** Data Preprocessing & Feature Engineering, Data Visualization (Matplotlib, Seaborn, Plotly), Pandas, NumPy
- **Soft Skills:** Problem Solving, Business & Product Understanding, Communication & Documentation